Log Book Entry 5.1

NB2420 Electronic Instrumentation · Manual 5.1 · Diode circuits

Hand in via BrightSpace

Name(s): Partner:

Date: Group:

PREPARATION

complete before the practicum

1. *What theories, laws, and/or formulas are most important for this experiment?*
2. *What do you expect to learn?*
3. *What do you expect to observe, and why?*

PREDICTION, SIMULATION, MEASUREMENT MATRIX

Task C1

Fill the conduction rule first, from Task B2. A row where prediction and measurement disagree is the useful row, so name the cause rather than smoothing it over.

Circuit	Conduction rule	Predicted feature	Simulated feature	Measured feature	Best explanation of mismatch
Series clipper, 1 V					
Shunt clipper, 1 V					
Shunt clipper, 5 V					
Biased clipper					
Silicon bridge				not built	
ALPACA LED bridge			not simulated		

Bridge and ripple Tasks A3, A4, S2, I4

Capacitor chosen, and the ripple it should give:

Ripple simulated:

Load polarity and pulse count observed:

Setup *Sketch one bridge current path from the source, through two elements, to the load and back. Mark which two conduct.*

GROUNDS AND CONCLUSIONS

Tasks A5, S3, I5, C1, C2

Shared grounds Tasks S3 and I5

What changed when the ground clip moved:

Device-model difference or grounding fault? How you told them apart:

C1 Reading the matrix *At which currents did the 0.7 V model predict your measurement well? The series and shunt clippers share a diode and a resistor, so name what decides the shape. Why do the LED bridge and the silicon bridge share a topology but not a waveform?*

C2 TA check-off *State one condition under which the 0.7 V model is adequate and one under which it is not, citing your own rows above.*

REFLECTION

complete before submission

1. *Which results matched or differed from your expectations, and why?*
2. *What is the most important thing you learned?*
3. *What would you do differently next time?*

TA check-off: Date: